

Zhuoyan Yu

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EDUCATION

New York University Shanghai
B.S. in Computer Science; GPA: 3.94

Expected May 2027
Shanghai, China

PUBLICATION

CodeFORGE: Framework for Orchestrated Rare Programming Language Generation and Evaluation

Zhuoyan Yu, Minghao Shao, Yijun Shen, Yichen Zhao, Boyuan Chen, Yik-Cheung Tam, Muhammad Shafique.
International Joint Conference on Neural Networks (IJCNN), 2026. [Paper] [Code]

RESEARCH EXPERIENCE

LLM-Generated Test Evaluation

Research Project, Serendipity Research Group

Jun 2026 – Present
NYU Shanghai

- Built a three-model evaluation pipeline comparing same-model and cross-model tests on fixed code candidates, with and without candidate visibility.
- Completed a study over 60 frozen LiveCodeBench problems, retaining 64 partially correct programs across 20 problems; implemented reference-consensus oracles, isolated execution, and reproducible analysis.

Ant Technologies U.S., Medical and Healthcare AI Lab

Algorithm Engineer Intern (Research)

May – Aug 2026

- Developed nnU-Net v2 pipelines for cerebral microbleed and lacune detection in multimodal MRI, including spatial alignment, data validation, and custom trainers.
- Improved microbleed lesion-level F1 from 0.533 to 0.557 on 72 VALDO cases using internal five-fold out-of-fold evaluation with cross-fitted thresholds.
- Conducted controlled ablations of voxel spacing, patch/batch size, loss, and foreground sampling; implemented lesion-level matching and free-response ROC analysis.

CodeFORGE: Rare and Legacy Code Generation

Research Assistant, Serendipity Research Fellowship

Sep 2025 – Apr 2026
NYU Shanghai

- Built a benchmark of 1,196 verified tasks across 12 rare and legacy languages, using a two-model pipeline that separates canonical solution synthesis from input/output test generation.
- Implemented Docker-based pass@k verification and evaluation with model cascading, feedback-driven resampling, and API cost controls.

Agentic Figure Transcription for Scientific Papers

Research Assistant, CSDSE Department

May – Aug 2025
NYU Shanghai

- Developed an academic-figure-to-editable-PPTX pipeline combining OCR, SAM segmentation, region-level VLM classification, and deterministic post-processing.

SELECTED PROJECT

MYGO: Geometry-Aware Depth ControlNet

Computer Vision Course Project

Sep – Dec 2025

- Implemented 18-channel depth conditioning for SD1.5 ControlNet, combining raw depth, 16 ordinal layers, and occlusion boundaries.
- Trained and compared baseline and MYGO models on 50k COCO images for 20k steps, evaluating image quality and depth/occlusion consistency.

SKILLS

Technical: Python, PyTorch, Hugging Face, OpenCV, Docker, Git.

Languages: Chinese, English.